

Dewen Lei

437-858-1566 | d24lei@uwaterloo.ca | linkedin.com/in/dewen-lei-1a3350290/ | github.com/DewenLei04

EDUCATION

University of Waterloo

Bachelor of Honours Mathematics, Major in Computational Mathematics

Minor in Computer Science

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Software Testing

Waterloo, ON

Aug. 2023 – Present

EXPERIENCE

Software Developer

Jan. 2026 – May. 2026

The EPCM Group

- Developed an **internal B2B quotation platform** for cathode and anode products, supporting quote pricing, proposal generation, and sales analytics for a product line with over **\$1M** in annual revenue.
- Modernized a legacy **MS Access** workflow into a **C#/.NET** internal tool, using Claude and Codex to accelerate development and reduce operational complexity.
- Built **C# automation scripts** to synchronize local SQL data with SAP through **OData APIs**, reducing repetitive manual updates and improving data consistency.
- Deployed an **AI chatbot workflow** for the company website to qualify leads, answer product questions, and support customer purchase decisions.

Software Developer

Jun. 2025 – Aug. 2025

China Telecom Cloud Computing

- Deduplicated and standardized a **30,000+ item question bank** using **Python** and **SentenceTransformers**.
- Designed the **UI/UX** for an academic **question-answering agent** based on client workflows, and authored a **client white paper** detailing **accuracy controls** and **onboarding steps**.
- Built a **labeling and quality loop** using an **LLM** to cross-check labels, flag **low-confidence** cases, and route edge cases for **human review**, improving precision and cutting review time.

PROJECTS

B2B SaaS Quotation Platform | *React, TypeScript, ASP.NET Core, MySQL*

Mar. 2026 – Apr. 2026

- Independently engineered and delivered a full-stack internal quotation portal for **cathode and anode products**, replacing a manual Excel-based pricing workflow with a standardized, database-driven web application.
- Developed the frontend with **React** and **TypeScript**, building parameter-driven quotation forms, editable proposal workflows, and analytics dashboards for sales and finance users.
- Built the backend with **ASP.NET Core** and **MySQL**, implementing **REST APIs** for quote calculation, customer records, pricing history, revision timelines, and proposal generation.
- Integrated **Microsoft Entra ID** authentication with **OAuth 2.0 Connect**, enabling secure employee sign-in through company Microsoft 365 accounts.
- Containerized services with **Docker** and deployed the application on **IIS**, supported by a **CI/CD pipeline** for repeatable production updates.

LLM-Powered Slide Generation Engine | *Vue, Python, LangChain, FastAPI*

Jun. 2025 – Jul. 2025

- Built a self-hosted AI backend using **FastAPI**, **LangChain**, and the **OpenAI API**, enabling real-time streaming and structured JSON output for automated slide generation.
- Developed a custom **binary search algorithm** for dynamic font-size calculation in AI-generated text elements, improving multilingual text fitting and rendering efficiency.
- Implemented indexed template matching with hash-based lookup tables, reducing template search complexity from **O(n²)** to **O(1)** for AI-generated slide content.
- Used **Postman** for API testing and validation, improving reliability, data integrity, and performance during backend development.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, C#, TypeScript, JavaScript, SQL, MATLAB, HTML/CSS, LaTeX

Frontend: React.js, Vue.js, HTML/CSS, jQuery

Backend: Node.js, ASP.NET Core, .NET, FastAPI, Spring Boot, REST APIs, LangChain

Databases: MySQL, MongoDB, Redis, SQL Server

Tools: Git, Docker, Linux, Agile, Postman, Microsoft Azure, SAP, IIS, Microsoft Access, Excel, PowerPoint

AI Coding Tools: Codex, Claude, Cursor, Copilot, Gemini